

Corrections with Safe Reversion Across Estimation Tasks:

What a Held-Out Check Can and Cannot Guarantee

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Abstract

Research line OMR001 proposed *post-estimation correction operators* learned across related estimation tasks, with the option to *revert to a safe reference* when a held-out check fails, and closed with the finding that sufficient differentiation from existing model combinations had not been demonstrated. The original operator definitions are not available; this note reconstructs a self-contained version and proves what such a construction can and cannot guarantee. In the Gaussian location model, (i) no correction operator, however it is fitted on source tasks whose law does not depend on the target parameter, improves the minimax risk of the task-specific estimator, so any gain must come from structure shared across tasks; in one or two dimensions every rule that ever uses a correction is strictly worse than the reference somewhere. (ii) If the check is a one-sided test at level α on the held-out loss difference computed from m independent observations, the risk of the reverting estimator exceeds the risk of the reference by at most $\min\{\alpha \mathbb{E}[\Delta^+], \varphi(z_{1-\alpha}) (2\sigma/\sqrt{m}) \mathbb{E}\|C - R\|\}$, where Δ is the loss increase the correction would cause, $C - R$ the size of the correction and φ the standard normal density; the probability of a harmful transfer event is at most α ; the bound is attained up to an explicit constant $\kappa(\alpha)/\varphi(z_{1-\alpha}) \in [0.05, 0.43]$ on the grid used, so the harm cannot be made $o(m^{-1/2})$ uniformly over operators; and the estimator is within $(z_{1-\alpha} + \varphi(0)) (2\sigma/\sqrt{m}) \mathbb{E}\|C - R\|$ of the better of the two candidates. (iii) The reverting estimator *is* penalised hold-out selection between two candidates, nothing in the guarantee uses that one candidate is a correction of the other, and the guarantee is the two-candidate case of standard hold-out bounds; the correction operator enters only through $\mathbb{E}\|C - R\|$ and Δ . This makes the closing statement of the line precise. A simulation with a fixed seed (345 configurations, 20 000 replicates each, 17 s of computation) confirms every bound, finds that with a predefined criterion the guaranteed construction is “useful” only when tasks are nearly identical because the held-out split costs 25.0% of the reference risk, and that the practical variants that avoid this cost (refit after the check; SURE-based reversion without a split) are classical constructs with no guarantee proved here. No general accuracy improvement and no transfer guarantee are claimed.

1 Introduction

Suppose an analyst estimates the same kind of quantity, a mean or a regression coefficient, in a family of related tasks. Each task has its own *reference* estimator, computed from that task’s data alone and trusted because it is unbiased or minimax. A *correction operator* maps the reference estimate to a corrected one using what was learned on the other tasks: shrinkage toward a pooled centre, an additive bias adjustment, a linear map fitted on source tasks. Since the correction can hurt when the target task does not share the structure, the proposal of research line OMR001 was to *revert* to the reference whenever a held-out check on the target task does not favour the correction. The public record of the line states its objective (evaluate corrections

on held-out tasks and establish conditions for transfer), its finding (sufficient differentiation from existing model combinations and operator-based corrections has not been demonstrated) and its scope (no general accuracy improvement or transfer guarantee is claimed).

The original definitions of the operators and of the “safe reference” are not available. This note therefore reconstructs a self-contained version of the construction (Section 2) and asks the question the line left open: *what, exactly, can a correction with safe reversion guarantee, and what can it not?* The answers are elementary but, we think, worth having in one place with proofs and with their constants.

1. **What cannot be guaranteed** (Section 3). No correction, fixed in advance or fitted on source tasks whose law does not depend on the target parameter, improves the minimax risk of the reference over all shifts of the mean; a fixed additive correction is uniformly worse by exactly its mean square; in one or two dimensions every rule that uses a correction with positive probability is strictly worse than the reference at some parameter value. Any gain must come from structure shared across tasks and is a statement about average risk over tasks, not about worst-case risk.
2. **What can be guaranteed** (Section 4). With a held-out sample of size m and a one-sided test at level α , the reverting estimator does no harm beyond an explicit amount of order $\sigma \mathbb{E}\|C - R\|/\sqrt{m}$, the probability of a harmful transfer event is at most α , and the estimator is within the same order of the better of the two candidates. The rate and the dependence on $\|C - R\|$ cannot be improved uniformly over operators.
3. **Why the line’s conclusion is right** (Section 5). The reverting estimator is hold-out selection between two candidates with a switching penalty. Its safety is a property of selection, not of the operator; its gain is a property of the shared structure, exploited by classical empirical Bayes operators. We list the constructs it coincides with or resembles.
4. **A simulation with a predefined criterion** (Section 6) checks the bounds, reports the frequency of harmful transfer and states honestly where “useful transfer” occurs.

Every mathematical statement below carries a status tag in Table 9: proved here, classical, from the literature, computationally verified, or empirical only. Every number in the text is generated from the results of one script with a fixed seed (Appendix A).

2 Setting and definitions

Tasks and data. A task is a parameter $\theta \in \mathbb{R}^d$ together with observations whose law depends on θ . We work throughout in the Gaussian location model with known variance: an observation of task θ is $Y \sim N(\theta, \sigma^2 I_d)$. The *target* task has parameter θ and provides $n = n_e + m$ independent observations; the first n_e form the *estimation sample*, with mean $\bar{X}_e \sim N(\theta, \sigma_e^2 I_d)$, $\sigma_e^2 = \sigma^2/n_e$, and the last m form the *held-out sample* $Y_1, \dots, Y_m$, with mean $\bar{Y} \sim N(\theta, \sigma^2 I_d/m)$. *Source* tasks $s = 1, \dots, S$ have parameters θ_s and produce data D whose law may or may not carry information about θ ; this is the whole question. We write $\mathcal{F}$ for the σ -field generated by the estimation sample and by D ; the held-out sample is independent of $\mathcal{F}$ given θ .

Loss and risk. For an estimate δ of θ the loss is $\ell(\delta) = \|\delta - \theta\|^2$ and the risk is $\mathcal{R}(\theta, \delta) = \mathbb{E}_\theta \ell(\delta)$. When θ is itself drawn from a distribution over tasks, the average of the risk is the Bayes risk.

Definition 2.1 (Reference, correction, reversion). (a) The *reference* estimator is the task-specific mean $R = \bar{X}_e$; it is unbiased and minimax for $\theta \in \mathbb{R}^d$ (Theorem 3.1(a)).

- (b) A *correction operator* is an $\mathcal{F}$ -measurable map producing a corrected estimate $C = C(R; D) \in \mathbb{R}^d$. Three instances are used below: the affine shrinkage $C = \hat{\mu} + \lambda(R - \hat{\mu})$ toward a centre $\hat{\mu}$ estimated from the sources, with λ either estimated from the sources (the empirical Bayes rule of Efron and Morris [6, 7], Morris [14]), fixed in advance, or equal to zero (full pooling).
- (c) The *held-out check* compares the held-out losses of the two candidates,

$$\hat{D} = \|C - \bar{Y}\|^2 - \|R - \bar{Y}\|^2,$$

and the *reverting estimator* with margin $c \geq 0$ ($\mathcal{F}$ -measurable) is

$$\hat{\theta}_{\text{rev}} = C \mathbf{1}\{\hat{D} \leq -c\} + R \mathbf{1}\{\hat{D} > -c\}.$$

The margin $c = z_{1-\alpha} s$ with $s = 2\sigma\|C - R\|/\sqrt{m}$ and $z_{1-\alpha} = \Phi^{-1}(1 - \alpha)$ makes the check a one-sided test at level α of “the correction does not help” (Lemma 4.1); $c = 0$ is plain hold-out selection.

We write $\Delta = \ell(C) - \ell(R)$ for the loss increase caused by the correction (negative when it helps), $\Delta^+ = \max(\Delta, 0)$, and $\pi = \mathbb{P}(\hat{D} \leq -c \mid \mathcal{F}, \theta)$ for the conditional probability that the check passes.

Shared structure. In the simulation the parameters of all tasks are exchangeable draws from $N(\mu, \tau^2 I_d)$ (a hierarchical or empirical Bayes model). The amount of shared structure is measured by $\text{snr} = \tau^2/\sigma_e^2$: at $\text{snr} = 0$ all tasks have the same parameter, at $\text{snr} \rightarrow \infty$ they share nothing. Departure from the structure is modelled by shifting the target’s centre by $\kappa\tau$ along one coordinate while the sources stay put.

3 What cannot be guaranteed

Theorem 3.1 (No free lunch for corrections). *Let $X \sim N(\theta, \sigma_e^2 I_d)$ with $\theta \in \mathbb{R}^d$ unknown and σ_e^2 known, and let D be auxiliary data, independent of X given θ , whose law Q does not depend on θ . Let $\delta = \delta(X, D)$ be any estimator.*

- (a) $\sup_{\theta} \mathcal{R}(\theta, \delta) \geq d\sigma_e^2 = \sup_{\theta} \mathcal{R}(\theta, X)$. [classical; proof included]
- (b) If $\delta = X - b(D)$ for a measurable b , then $\mathcal{R}(\theta, \delta) = d\sigma_e^2 + \mathbb{E}\|b(D)\|^2$ for every θ . [proved here]
- (c) If $d \leq 2$ and $\mathbb{P}_{\theta}(\delta \neq X) > 0$ for some (equivalently every) θ , then $\mathcal{R}(\theta', \delta) > d\sigma_e^2$ for some θ' . [literature (admissibility) + proved here]
- (d) If $d \geq 3$ there exist δ with $\mathcal{R}(\theta, \delta) < d\sigma_e^2$ for every θ ; by (a) the supremum is still $d\sigma_e^2$. [literature]

Proof. (a) Fix $\tau > 0$ and put the prior $\theta \sim N(0, \tau^2 I_d)$, so that (θ, X, D) has joint law $\pi(d\theta)P_{\theta}(dx)Q(dD)$. Because Q does not depend on θ and D is independent of X given θ , the conditional law of θ given (X, D) equals its conditional law given X , namely $N(\frac{\tau^2}{\tau^2 + \sigma_e^2} X, \frac{\tau^2 \sigma_e^2}{\tau^2 + \sigma_e^2} I_d)$. The Bayes risk of any $\delta(X, D)$ is therefore at least that of the posterior mean, $d\tau^2 \sigma_e^2 / (\tau^2 + \sigma_e^2)$. Since the supremum of the risk dominates its average, $\sup_{\theta} \mathcal{R}(\theta, \delta) \geq d\tau^2 \sigma_e^2 / (\tau^2 + \sigma_e^2)$ for every τ ; let $\tau \rightarrow \infty$. Equality for $\delta = X$ is direct. (b) $\mathbb{E}_{\theta}\|X - b - \theta\|^2 = \mathbb{E}\|X - \theta\|^2 + \mathbb{E}\|b\|^2 - 2\mathbb{E}\langle X - \theta, b \rangle$, and the cross term vanishes because $X - \theta$ has mean zero and is independent of D . (c) Let $\tilde{\delta}(X) = \mathbb{E}[\delta(X, D) \mid X]$, which does not depend on θ because the conditional law of D given X is Q . By the parallelogram identity, $\mathcal{R}(\theta, \delta) = \mathcal{R}(\theta, \tilde{\delta}) + \mathbb{E}_{\theta}\|\delta - \tilde{\delta}\|^2$. If $\mathbb{P}(\tilde{\delta} \neq X) > 0$, suppose for contradiction that $\mathcal{R}(\theta, \tilde{\delta}) \leq d\sigma_e^2$ for all θ ; then $\delta' = (\tilde{\delta} + X)/2$ satisfies $\|\delta' - \theta\|^2 = \frac{1}{2}\|\tilde{\delta} - \theta\|^2 + \frac{1}{2}\|X - \theta\|^2 - \frac{1}{4}\|\tilde{\delta} - X\|^2$, hence $\mathcal{R}(\theta, \delta') \leq d\sigma_e^2 - \frac{1}{4}\mathbb{E}_{\theta}\|\tilde{\delta} - X\|^2 < d\sigma_e^2$ for every θ

(the Gaussian laws P_θ are mutually absolutely continuous, so the last expectation is positive for all θ once it is positive for one). This contradicts the admissibility of X for $d \leq 2$ [3, 16]; see Lehmann and Casella [11, Ch. 5]. If instead $\tilde{\delta} = X$ a.s., then $\mathcal{R}(\theta, \delta) = d\sigma_e^2 + \mathbb{E}_\theta \|\delta - X\|^2 > d\sigma_e^2$ for every θ . (d) is the James–Stein theorem [10, 16]; the risk of the James–Stein estimator tends to $d\sigma_e^2$ as $\|\theta\| \rightarrow \infty$. $\square$

Remark 3.2 (Where transfer can enter). The only hypothesis of Theorem 3.1(a) that a transfer setting can violate is that the law of the source data does not depend on θ . When θ and the source parameters are exchangeable draws from $N(\mu, \tau^2 I_d)$, the sources are informative about θ through μ and τ^2 , and the natural criterion is the Bayes risk over tasks. For the affine correction $C = \mu + \lambda(X - \mu)$ with μ and $\lambda \in [0, 1)$ fixed,

$$\mathcal{R}(\theta, C) = \lambda^2 d\sigma_e^2 + (1 - \lambda)^2 \|\theta - \mu\|^2, \quad \mathbb{E}_{\theta \sim N(\mu, \tau^2 I)} \mathcal{R}(\theta, C) = \lambda^2 d\sigma_e^2 + (1 - \lambda)^2 d\tau^2, \quad (1)$$

minimised at $\lambda^* = \tau^2 / (\tau^2 + \sigma_e^2)$ with value $d\sigma_e^2 \lambda^* < d\sigma_e^2$. [classical] The available relative gain over the reference is $1 - \lambda^* = 1/(1 + \text{snr})$. The first formula in (1) is the negative-transfer risk: at a target far from the centre the correction is worse than the reference by $(1 - \lambda)^2 \|\theta - \mu\|^2 - (1 - \lambda^2) d\sigma_e^2$, unbounded in θ . Reversion is meant to cap this term; Section 4 says by how much.

4 What can be guaranteed

Lemma 4.1 (The held-out loss difference). *With the notation of Definition 2.1, $\hat{D} - \Delta = -2\langle C - R, \bar{Y} - \theta \rangle$. Hence $\mathbb{E}[\hat{D} \mid \mathcal{F}, \theta] = \Delta$, and if the held-out observations are Gaussian then, conditionally on $(\mathcal{F}, \theta)$, $\hat{D} \sim N(\Delta, s^2)$ with $s = 2\sigma\|C - R\|/\sqrt{m}$. In particular the rule “use C iff $\hat{D} \leq -z_{1-\alpha}s$ ” rejects the hypothesis $\Delta \geq 0$ with conditional probability $\Phi(-(\Delta + z_{1-\alpha}s)/s) \leq \alpha$ whenever $\Delta \geq 0$.*

Proof. Expand $\|C - \bar{Y}\|^2 - \|R - \bar{Y}\|^2 = \|C\|^2 - \|R\|^2 - 2\langle C - R, \bar{Y} \rangle$ and $\Delta = \|C\|^2 - \|R\|^2 - 2\langle C - R, \theta \rangle$. Given $(\mathcal{F}, \theta)$, $\bar{Y} - \theta \sim N(0, \sigma^2 I_d/m)$ is independent of $C - R$, so $\langle C - R, \bar{Y} - \theta \rangle \sim N(0, \sigma^2 \|C - R\|^2/m)$. $\square$

Theorem 4.2 (Do no harm, up to a controlled amount). *Let R, C be $\mathcal{F}$ -measurable, let the held-out sample be independent of $\mathcal{F}$ given θ with mean θ , and let $\hat{\theta}_{\text{rev}}$ be the reverting estimator with margin $c \geq 0$. Write $s = 2\sigma\|C - R\|/\sqrt{m}$.*

(i) (Identity.) $\mathcal{R}(\hat{\theta}_{\text{rev}}) - \mathcal{R}(R) = \mathbb{E}[\Delta \pi] \leq \mathbb{E}[\Delta^+ \pi]$, where $\pi = \mathbb{P}(\hat{D} \leq -c \mid \mathcal{F}, \theta)$.

(ii) (Gaussian held-out, test at level $\alpha \in (0, \frac{1}{2}]$, $c = z_{1-\alpha}s$.)

$$\mathcal{R}(\hat{\theta}_{\text{rev}}) - \mathcal{R}(R) \leq \min \left\{ \alpha \mathbb{E}[\Delta^+], \quad \varphi(z_{1-\alpha}) \frac{2\sigma}{\sqrt{m}} \mathbb{E}\|C - R\| \right\},$$

and the probability of a harmful transfer event satisfies $\mathbb{P}(\hat{\theta}_{\text{rev}} = C, \Delta > 0) \leq \alpha \mathbb{P}(\Delta > 0) \leq \alpha$.

(iii) (Bounded held-out.) *If instead $\|Y_j - \theta\| \leq \rho$ a.s. and $c = 2\rho\|C - R\|\sqrt{2\log(1/\alpha)/m}$, then the same two statements hold with $\varphi(z_{1-\alpha})2\sigma/\sqrt{m}$ replaced by $2\rho/\sqrt{e m}$.*

(iv) (Near-oracle selection, Gaussian held-out, $c = z_{1-\alpha}s$.)

$$\mathcal{R}(\hat{\theta}_{\text{rev}}) \leq \mathbb{E}[\min\{\ell(R), \ell(C)\}] + (z_{1-\alpha} + \varphi(0)) \frac{2\sigma}{\sqrt{m}} \mathbb{E}\|C - R\|.$$

[proved here; standard arguments]

Proof. (i) $\ell(\hat{\theta}_{\text{rev}}) = \ell(R) + \Delta \mathbf{1}\{\hat{D} \leq -c\}$. Take expectations conditionally on $(\mathcal{F}, \theta)$, on which Δ is measurable: $\mathbb{E}[\ell(\hat{\theta}_{\text{rev}}) - \ell(R) \mid \mathcal{F}, \theta] = \Delta\pi$. On $\{\Delta \leq 0\}$ this is ≤ 0 , whence the inequality.

(ii) By Lemma 4.1, on $\{\Delta > 0\}$ we have $\pi = \Phi(-(z+u))$ with $z = z_{1-\alpha} \geq 0$ and $u = \Delta/s \geq 0$ (if $s = 0$ then $C = R$, $\Delta = 0$ and there is nothing to prove). First bound: $\Phi(-(z+u)) \leq \Phi(-z) = \alpha$, so $\mathbb{E}[\Delta^+\pi] \leq \alpha\mathbb{E}[\Delta^+]$. Second bound: by the Mills ratio inequality $\Phi(-x) \leq \varphi(x)/x$ for $x > 0$,

$$\Delta^+\pi = s u \Phi(-(z+u)) \leq s \frac{u}{z+u} \varphi(z+u) \leq s \varphi(z+u) \leq s \varphi(z),$$

using that φ decreases on $[0, \infty)$. Taking expectations gives $\varphi(z)\mathbb{E}[s]$. For the harmful event, $\mathbb{P}(\hat{\theta}_{\text{rev}} = C, \Delta > 0) = \mathbb{E}[\mathbf{1}\{\Delta > 0\}\pi] \leq \alpha\mathbb{P}(\Delta > 0)$.

(iii) Write $\hat{D} - \Delta = \frac{1}{m} \sum_j W_j$ with $W_j = -2\langle C - R, Y_j - \theta \rangle$, independent given $(\mathcal{F}, \theta)$, mean zero and $|W_j| \leq B := 2\rho\|C - R\|$. Hoeffding's inequality [9] gives $\mathbb{P}(\hat{D} - \Delta \leq -t \mid \mathcal{F}, \theta) \leq \exp(-mt^2/(2B^2))$. On $\{\Delta \geq 0\}$, $\pi \leq \exp(-m(c+\Delta)^2/(2B^2)) \leq \exp(-mc^2/(2B^2)) = \alpha$ for the stated c , which yields the α -bound and the bound on the harmful event. Also $\Delta^+\pi \leq \Delta^+ \exp(-m(\Delta^+)^2/(2B^2)) \leq \sup_{x \geq 0} x e^{-mx^2/(2B^2)} = B/\sqrt{em}$.

(iv) $\ell(\hat{\theta}_{\text{rev}}) - \min\{\ell(R), \ell(C)\} = |\Delta| \mathbf{1}\{\text{wrong choice}\}$, where the choice is wrong on $\{\Delta > 0, \hat{D} \leq -c\}$ and on $\{\Delta < 0, \hat{D} > -c\}$. Conditionally on $(\mathcal{F}, \theta)$, with $u = |\Delta|/s$: on $\{\Delta > 0\}$ the conditional expectation is $s u \Phi(-(z+u)) \leq s \varphi(z+u) \leq s \varphi(0)$ as in (ii); on $\{\Delta < 0\}$ it is $s u \Phi(z-u)$ by Lemma 4.1, which is at most $s z$ when $u \leq z$ and, writing $u = z + v$ with $v > 0$, at most $s(z + v\Phi(-v)) \leq s(z + \varphi(v)) \leq s(z + \varphi(0))$ otherwise. Both cases are bounded by $s(z + \varphi(0))$; take expectations. $\square$

Corollary 4.3 (Price of the split). *Let $R_n = \bar{X}_n$ be the reference computed from all $n = n_e + m$ target observations, with risk $d\sigma^2/n$. Then*

$$\mathcal{R}(\hat{\theta}_{\text{rev}}) - \mathcal{R}(R_n) = d\sigma^2 \left(\frac{1}{n_e} - \frac{1}{n} \right) + (\mathcal{R}(\hat{\theta}_{\text{rev}}) - \mathcal{R}(R)),$$

so the total cost of the safety mechanism relative to “do nothing with all the data” is the split cost $d\sigma^2 m/(nn_e)$, a fraction m/n_e of the reference risk, plus the controlled term of Theorem 4.2. [proved here (arithmetic)]

Proposition 4.4 (Sharpness of the constant). *For $\alpha \in (0, \frac{1}{2}]$ let $\kappa(\alpha) = \sup_{u \geq 0} u \Phi(-(z_{1-\alpha} + u))$. Then $\kappa(\alpha) \leq \varphi(z_{1-\alpha})$ and, for every $\varepsilon > 0$, there is a Gaussian target model and an $\mathcal{F}$ -measurable pair (R, C) with*

$$\mathcal{R}(\hat{\theta}_{\text{rev}}) - \mathcal{R}(R) \geq (\kappa(\alpha) - \varepsilon) \frac{2\sigma}{\sqrt{m}} \mathbb{E}\|C - R\|.$$

Consequently the second bound in Theorem 4.2(ii) cannot be improved beyond the factor $\varphi(z_{1-\alpha})/\kappa(\alpha)$, and no held-out check of size m makes the worst-case harm $o(\sigma\mathbb{E}\|C - R\|/\sqrt{m})$ uniformly over correction operators. [proved here; constants computationally verified, Table 8]

Proof. The inequality $\kappa \leq \varphi(z)$ is the chain displayed in the proof of Theorem 4.2(ii). For the lower bound take $R = \bar{X}_e$ and $C = R + v$ with a fixed vector $v \neq 0$ (an additive correction fixed in advance, as in Theorem 3.1(b)). Then $\|C - R\| = \|v\|$, $s = 2\sigma\|v\|/\sqrt{m}$ is deterministic and $\Delta = \|v\|^2 + 2\langle v, R - \theta \rangle$. By Theorem 4.2(i) the excess risk equals $\mathbb{E}[g(\Delta)]$ with $g(x) = x \Phi(-(x + zs)/s)$, a continuous function with $|g(x)| \leq |x|$. As $n_e \rightarrow \infty$, $\Delta \rightarrow \|v\|^2$ in L^2 , so $\mathbb{E}[g(\Delta)] \rightarrow g(\|v\|^2) = s u \Phi(-(z+u))$ with $u = \|v\|^2/s = \|v\|\sqrt{m}/(2\sigma)$, which can be given any prescribed value $u \geq 0$ by the choice of $\|v\|$. Choosing u within $\varepsilon/2$ of the supremum and n_e large gives the claim. $\square$

Remark 4.5 (Detectability, not estimability). When the correction helps ($\Delta < 0$), Lemma 4.1 gives the conditional probability of using it, $\Phi(-z_{1-\alpha} + |\Delta|/s)$ with $|\Delta|/s = |\Delta|\sqrt{m}/(2\sigma\|C - R\|)$.

In the shift example of Proposition 4.4 with a beneficial v this ratio is $\|v\|\sqrt{m}/(2\sigma)$ whatever the dimension: the check is a one-dimensional test along the direction $C - R$ supplied by the sources, so a gain can be *verified* on the target with a held-out sample that would be far too small to *learn* the correction there. Gains too small to be detected at level α are forfeited, up to the regret term of Theorem 4.2(iv).

Remark 4.6 (Variants of the check). (a) If σ is unknown, the check can be the one-sided t -test at level α on the m scalar projections $\langle C - R, Y_j \rangle$; under Gaussian held-out observations it is exact at the boundary $\Delta = 0$ and its power is monotone in the non-centrality [12], so the α -form of Theorem 4.2(ii) and the bound on the harmful event hold verbatim. (b) With only second moments ($\text{Var}(\langle C - R, Y_j - \theta \rangle) \leq \sigma^2 \|C - R\|^2$) and $c = zs$ with $z > 0$, Chebyshev’s inequality and $(c + \Delta)^2 \geq 4c\Delta$ give $\mathcal{R}(\hat{\theta}_{\text{rev}}) - \mathcal{R}(R) \leq \mathbb{E}[s]/(4z)$, the same order in m with a worse constant. (c) None of the statements uses that C is a function of R : they hold for any two $\mathcal{F}$ -measurable candidates.

5 Relation to known constructs, and the line’s conclusion

Proposition 5.1 (The reverting estimator is penalised hold-out selection). *Let $\hat{L}_m(\delta) = \frac{1}{m} \sum_{j=1}^m \|\delta - Y_j\|^2$ be the held-out empirical loss. Then $\hat{L}_m(C) - \hat{L}_m(R) = \hat{D}$, and*

$$\hat{\theta}_{\text{rev}} = \arg \min_{\delta \in \{R, C\}} \{ \hat{L}_m(\delta) + c \mathbf{1}\{\delta = C\} \}$$

(ties resolved toward R). With $c = 0$ it is plain hold-out selection between two candidates. [proved here (one line)]

Proof. $\hat{L}_m(\delta) = \|\delta - \bar{Y}\|^2 + \frac{1}{m} \sum_j \|Y_j - \bar{Y}\|^2$, and the second term does not depend on δ . $\square$

Three consequences follow. First, the guarantee of Theorem 4.2 is the two-candidate case of the standard analysis of hold-out selection among finitely many candidates fixed before the held-out sample is seen [4, Ch. 8], see also Arlot and Celisse [1]; with K candidates the same argument costs an extra $\sqrt{\log K}$, and Theorem 4.2 only sharpens the constants in the Gaussian two-candidate case and makes the one-sided, “do no harm” reading explicit. Second, the correction operator enters the guarantee only through $\mathbb{E}\|C - R\|$, the size of the proposed change, and through Δ , its actual effect; the guarantee is indifferent to how the operator was obtained. Third, whether the reverting estimator *gains* anything is entirely a question about the shared structure (Remark 3.2) and about whether that gain is detectable with m held-out points (Remark 4.5). Table 1 places the construction next to the classical ones it resembles.

Why the line’s conclusion is correct. The public finding of OMR001 was that sufficient differentiation from existing model combinations and operator-based corrections had not been demonstrated. Proposition 5.1 and Theorem 4.2 say why no such differentiation should be expected from the safety mechanism: safe reversion *is* held-out selection between two candidates, its only guarantee is a selection guarantee, and that guarantee is available for any pair of candidates. Differentiation could only come from the operator family itself, and the natural operators (shrinkage toward a pooled centre, additive bias adjustments) are the empirical Bayes rules of Efron and Morris [6]. We therefore regard the closing statement of the line as established rather than merely prudent. What remains of value is the explicit form of the guarantee, its sharpness in m and $\|C - R\|$, and the negative finding of the next section about the price of the split.

construct	reverts to	trigger	guarantee
Positive-part James–Stein [2]	the shrinkage target	$\ X - \mu\ ^2 < (d - 2)\sigma_e^2$	dominates James–Stein; minimax, $d \geq 3$
Limited-translation rules [5]	the reference, coordinate-wise	component moved further than a cap	bounds the risk increase of each component
SURE selection [17]	either candidate	sign of an unbiased risk-difference estimate; no split	unbiased estimate, no finite-sample selection bound in general
Hold-out / cross-validated selection [1, 4]	either candidate	held-out loss difference	regret $O(\text{range}/\sqrt{m})$
Source detection in transfer learning [13]	target-only estimator	data-splitting aggregation on the target	rates under informative sources; protection against non-informative ones
Reverting estimator (this note)	the reference R	one-sided test at level α on $\hat{D}$	Theorem 4.2: $\text{harm} \leq \min\{\alpha\mathbb{E}\Delta^+, \varphi(z)2\sigma\mathbb{E}\ C - R\ /\sqrt{m}\}$

Table 1: The reverting estimator next to constructs it coincides with or resembles. The first two revert in a different direction or by a different mechanism; the third and fourth are the same construct with a different check; “negative transfer” is the name given to $\Delta > 0$ in the transfer-learning literature [15].

6 Simulation

Protocol. $d = 5$, $\sigma = 1$, $S = 30$ source tasks with $n_s = 60$ observations each, a target task with $n = 60$ observations split into $n_e = 48$ for estimation and $m = 12$ for the check (a 25.0% split cost, Corollary 4.3); $m \in \{3, 6, 12, 24\}$ is also swept. Sources and target are exchangeable draws from $N(\mu, \tau^2 I_d)$ with $\text{snr} = \tau^2/\sigma_e^2 \in \{0, 0.25, 0.5, 1, 2, 4, 8, 16\}$ (structure sweep), and, at $\text{snr} = 1$, the target centre is shifted by $\kappa\tau$ along one coordinate with $\kappa \in \{0, 2, 4, 6, 8, 12, 16\}$ (departure sweep). The operators are the empirical Bayes shrinkage $C = \hat{\mu} + \hat{\lambda}(R - \hat{\mu})$ with $\hat{\mu}$ the mean of the source means and $\hat{\lambda} = \hat{\tau}^2/(\hat{\tau}^2 + \sigma_e^2)$, $\hat{\tau}^2 = (\hat{v} - \sigma^2/n_s)^+$, $\hat{v}$ the pooled variance of the source means; full pooling ($\lambda = 0$); and a fixed $\lambda = \frac{1}{2}$. The check uses $\alpha \in \{0.5, 0.2, 0.1, 0.05, 0.01\}$. Each configuration uses 20 000 replicates with common random numbers across α and operators, seed 20260930. Two variants *without* a guarantee proved here are also run: *refit*, which applies the same check but then builds the final estimate from all n observations, and *SURE reversion*, which uses no split and applies C to $\bar{X}_n$ iff Stein’s unbiased estimate [17] of $\mathcal{R}(C) - \mathcal{R}(\bar{X}_n)$, equal to $(1 - \hat{\lambda}_n)[(1 - \hat{\lambda}_n)\|\bar{X}_n - \hat{\mu}\|^2 - 2d\sigma^2/n]$ for the affine operator, is negative.

Predefined criteria. (U) *Useful transfer* at a configuration: the lower end of the paired 95% confidence interval of the relative risk gain with respect to the *full-data* reference $\bar{X}_n$ is at least 5.0%; the split is charged to the method. (S) *Safety*: the Monte Carlo excess risk of the reverting estimator over R never exceeds any of the three bounds of Theorem 4.2(i)–(ii) by more than two Monte Carlo standard errors, and the harmful-event frequency never exceeds $\alpha \hat{\mathbb{P}}(\Delta > 0)$ by more than two binomial standard errors. (H) The frequency of harmful transfer events, $\hat{\theta}_{\text{rev}} = C$ and $\Delta > 0$, is reported for every configuration.

Results: safety (S). Across all 345 (configuration, operator, α) combinations the three bounds hold with the two-standard-error allowance: yes; the harmful-event frequency bound holds: yes. The largest ratio of the Monte Carlo excess to the closed-form bound $\varphi(z)\mathbb{E}[s]$ is 0.42, and to the regret bound of Theorem 4.2(iv) it is 0.42. The identity of Theorem 4.2(i) was checked replicate by replicate: where acceptance is frequent (273 combinations) the paired z -statistic has maximum modulus 2.96; where it is rare (72 combinations) the acceptance count

snr	λ^*	$\bar{X}_n$	always	reverting, with guarantee				without guarantee	
			all data	$\alpha=0.5$	$\alpha=0.2$	$\alpha=0.1$	$\alpha=0.01$	refit 0.1	SURE
0	0.00	0.0833	0.0030 ^U	0.0295 ^U	0.0610 ^U	0.0769 ^U	0.0996	0.0702 ^U	0.0176 ^U
0.25	0.20	0.0833	0.0233 ^U	0.0453 ^U	0.0697 ^U	0.0825	0.0998	0.0743 ^U	0.0365 ^U
0.5	0.33	0.0831	0.0347 ^U	0.0542 ^U	0.0745 ^U	0.0853	0.1002	0.0759 ^U	0.0449 ^U
1	0.50	0.0841	0.0485 ^U	0.0660 ^U	0.0815	0.0897	0.1015	0.0789 ^U	0.0562 ^U
2	0.67	0.0832	0.0605 ^U	0.0773 ^U	0.0877	0.0933	0.1015	0.0801	0.0653 ^U
4	0.80	0.0833	0.0699 ^U	0.0868	0.0935	0.0973	0.1033	0.0813	0.0728 ^U
8	0.89	0.0833	0.0761 ^U	0.0933	0.0973	0.0997	0.1034	0.0824	0.0776 ^U
16	0.94	0.0833	0.0797	0.0976	0.0998	0.1014	0.1037	0.0828	0.0804

Table 2: Structure sweep, EB operator, target drawn from the hierarchy, $m = 12$. Risks (mean squared error over $d = 5$ coordinates; Monte Carlo standard errors are below 0.001). ^U marks configurations meeting the predefined criterion U (gain $\geq 5.0\%$ over the full-data reference with 95% confidence). The reference on the estimation sample has risk ≈ 0.1042 against 0.0833 for $\bar{X}_n$.

snr	R	C (split)	$\hat{\mathbb{P}}(\Delta > 0)$ %	rev 0.1	harm %	excess	bound (i)	bound φ	bound α
0	0.1044	0.0030	0.1	0.0769	0.0	-0.0275	0.0000	0.0306	0.0000
0.25	0.1042	0.0243	5.2	0.0825	0.4	-0.0217	0.0000	0.0284	0.0001
0.5	0.1041	0.0376	10.4	0.0853	0.7	-0.0188	0.0001	0.0260	0.0002
1	0.1048	0.0545	17.7	0.0897	1.3	-0.0152	0.0002	0.0225	0.0004
2	0.1039	0.0709	24.9	0.0933	1.6	-0.0106	0.0003	0.0184	0.0006
4	0.1052	0.0846	30.7	0.0973	1.8	-0.0079	0.0003	0.0143	0.0007
8	0.1046	0.0934	36.1	0.0997	2.3	-0.0049	0.0003	0.0107	0.0007
16	0.1044	0.0987	40.6	0.1014	2.4	-0.0030	0.0002	0.0078	0.0006

Table 3: Theorem 4.2 on the structure sweep, $\alpha = 0.1$: risks of the reference $R = \bar{X}_e$ and of the corrected estimate on the same sample, frequency of $\Delta > 0$, risk and harmful-event frequency of the reverting estimator, its Monte Carlo excess over R , and the three bounds $\mathbb{E}[\Delta^+ \pi]$, $\varphi(z)\mathbb{E}[s]$, $\alpha\mathbb{E}[\Delta^+]$.

is compared with $\sum \pi$ through a Poisson 99% interval, and 2 combinations fall outside it, both from the same random batch (full pooling, $\alpha = 0.05$, $m = 6$, departure 8τ , 26 acceptances against 48 expected). Re-simulating that configuration with 50 independent seeds gives 2488 acceptances against 2426 expected ($z = +1.27$), and a direct conditional check with one fixed triple (R, C, θ) and fresh held-out draws agrees with $\Phi(-(\Delta + zs)/s)$ at every α (largest $|z|$ 1.72). The harmful-event frequency of the EB operator never exceeds 25.3% at $\alpha = 0.5$, 3.4% at $\alpha = 0.1$ and 0.2% at $\alpha = 0.01$, against 30.7%–100.0% for always correcting on the split sample (Tables 3–5). Table 7 shows the same cap for full pooling, whose unreverted risk reaches 1.7324 at snr = 16 against 0.1064 after reversion.

Results: usefulness (U). Always correcting with all the data is useful at $\text{snr} \in \{0, 0.25, 0.5, 1, 2, 4, 8\}$ and, in the hierarchy, never harmful in the criterion’s sense; its relative gain over $\bar{X}_n$ is +42.3% at $\text{snr} = 1$. The reverting estimator *with* the guarantee is useful at $\text{snr} \in \{0, 0.25, 0.5, 1, 2\}$ for $\alpha = 0.5$, $\{0, 0.25, 0.5\}$ for $\alpha = 0.2$, $\{0\}$ for $\alpha = 0.1$, and nowhere for $\alpha \leq 0.05$: at $\text{snr} = 1$ its gain over $\bar{X}_n$ is +21.5% ($\alpha = 0.5$), -6.6% ($\alpha = 0.1$) and -20.8% ($\alpha = 0.01$), while its gain over the reference on the same estimation sample is +37.1%, +14.5% and +3.1%. The difference is the 25.0% split cost of Corollary 4.3, which the guarantee does not cover, plus the gain forfeited by the test (Remark 4.5): at $\text{snr} = 1$ and $\alpha = 0.1$ the correction is used in 19.6% of the replicates although it helps in 1 – 17.7% of them. Table 6 shows the trade-off in m : with $m = 3$ (a 5% split) the guaranteed estimator gains +3.6% at $\kappa = 0$ and loses -18.0% at $\kappa = 8$ with excess +0.0104 over R (bound 0.1155); with $m = 24$ it loses -29.2% and -68.4%. The two variants without guarantee are useful over a wider range: refit at $\text{snr} \in \{0, 0.25, 0.5, 1\}$ ($\alpha = 0.1$)

κ	$\bar{X}_n$	always	reverting, with guarantee				without guarantee	
		all data	$\alpha=0.5$	$\alpha=0.2$	$\alpha=0.1$	$\alpha=0.01$	refit 0.1	SURE
0	0.0837	0.0481	0.0655	0.0814	0.0895	0.1012	0.0785	0.0557
2	0.0842	0.0652	0.0792	0.0891	0.0947	0.1025	0.0822	0.0747
4	0.0831	0.1174	0.1074	0.1021	0.1017	0.1031	0.0859	0.0939
6	0.0837	0.2024	0.1340	0.1112	0.1070	0.1048	0.0888	0.0869
8	0.0834	0.3225	0.1528	0.1160	0.1084	0.1043	0.0890	0.0835
12	0.0838	0.6665	0.1621	0.1153	0.1085	0.1050	0.0877	0.0838
16	0.0835	1.1512	0.1454	0.1098	0.1054	0.1042	0.0847	0.0835

Table 4: Departure sweep, EB operator, $\text{snr} = 1$, target centre shifted by $\kappa\tau$, $m = 12$. Risks as in Table 2.

κ	R	C (split)	$\hat{\mathbb{P}}(\Delta > 0)$ %	harm %	use C $\Delta > 0$ %	excess $\pm$ SE	bound (i)	bound φ	bound α
0	0.1044	0.0542	17.2	1.3	7.7	-0.0148 ± 0.0003	0.0002	0.0226	0.0004
2	0.1045	0.0755	33.3	2.2	6.7	-0.0097 ± 0.0003	0.0006	0.0268	0.0012
4	0.1036	0.1413	68.8	3.4	4.9	-0.0019 ± 0.0002	0.0020	0.0368	0.0056
6	0.1047	0.2481	90.2	3.2	3.5	$+0.0023 \pm 0.0002$	0.0037	0.0493	0.0149
8	0.1041	0.3992	97.8	2.2	2.3	$+0.0042 \pm 0.0002$	0.0044	0.0629	0.0296
12	0.1049	0.8303	99.9	0.7	0.7	$+0.0035 \pm 0.0003$	0.0034	0.0914	0.0725
16	0.1042	1.4383	100.0	0.1	0.1	$+0.0012 \pm 0.0002$	0.0015	0.1204	0.1334

Table 5: Theorem 4.2 on the departure sweep, $\alpha = 0.1$. The conditional frequency of using C when it is harmful never exceeds α (Theorem 4.2(ii)).

and SURE reversion at $\text{snr} \in \{0, 0.25, 0.5, 1, 2, 4, 8\}$, the latter within a few points of always correcting in the hierarchy.

Results: negative transfer. On the departure sweep always correcting with all the data becomes harmful (upper end of the 95% interval of the gain below zero) at $\kappa \in \{4, 6, 8, 12, 16\}$ and reaches 13.8 times the reference risk at $\kappa = 16$. The guaranteed reverting estimator at $\alpha = 0.1$ stays between 1.07 and 1.30 times the full-data reference over the whole sweep; its excess over R is $+0.0042$ at $\kappa = 8$ against the bounds 0.0044 (identity), 0.0629 (φ) and 0.0296 (α), and 0.0012 at $\kappa = 16$ against 0.1204 and 0.1334. The unguaranteed variants are also capped in this design (SURE reversion is harmful only at $\kappa \in \{4, 6, 8\}$, by at most 1.13 times the reference), but nothing proved here says they must be.

Verdict under the predefined criterion. Safety holds everywhere, as the theorem says it must. Useful transfer with the guaranteed construction occurs only when the tasks are nearly identical ($\text{snr} \leq 2$ with plain hold-out, $\text{snr} = 0$ at $\alpha = 0.1$) because the held-out split costs more than the transfer gains elsewhere. The constructions that avoid the split and are useful over a wide range are the classical ones (empirical Bayes shrinkage with SURE selection; cross-validated selection followed by a refit), for which no finite-sample “do no harm” bound is proved here.

7 Claims and their status

Limitations. The model is Gaussian location with known variance; the identity and the φ -bound use Gaussian held-out observations, the Hoeffding version needs bounded ones, and the t -test variant is only remarked on. The reference is the sample mean; for a biased reference (a shared instrument bias, say) the held-out sample of the target carries the same bias and the check cannot see the correction, so that setting needs unbiased calibration data and is not covered. The simulation has one design ($d = 5$, $n = 60$, $S = 30$) and one structural family;

m	n_e	κ	R	$\bar{X}_n$	always	rev 0.1	gain	excess	bound φ	refit	SURE
3	57	0	0.0874	0.0830	0.0448	0.0800	+3.6%	-0.0074	0.0416	0.0785	0.0535
3	57	8	0.0872	0.0827	0.3229	0.0976	-18.0%	+0.0104	0.1155	0.0945	0.0830
6	54	0	0.0926	0.0837	0.0457	0.0830	+0.8%	-0.0096	0.0302	0.0790	0.0539
6	54	8	0.0923	0.0830	0.3227	0.0994	-19.7%	+0.0070	0.0838	0.0914	0.0832
12	48	0	0.1039	0.0832	0.0477	0.0886	-6.5%	-0.0153	0.0226	0.0779	0.0554
12	48	8	0.1040	0.0835	0.3241	0.1085	-29.9%	+0.0045	0.0631	0.0895	0.0836
24	36	0	0.1392	0.0833	0.0535	0.1076	-29.2%	-0.0316	0.0183	0.0774	0.0600
24	36	8	0.1380	0.0830	0.3144	0.1397	-68.4%	+0.0017	0.0513	0.0858	0.0830

Table 6: Held-out size, EB operator, $\text{snr} = 1$, $\alpha = 0.1$, $n = 60$ fixed. “gain” is relative to $\bar{X}_n$.

snr	operator	C (split)	$\hat{\mathbb{P}}(\Delta > 0)$ %	rev 0.1	harm %	excess	bound φ	bound α
0	EB shrinkage	0.0030	0.1	0.0769	0.0	-0.0275	0.0306	0.0000
0	full pooling	0.0028	0.1	0.0776	0.0	-0.0268	0.0316	0.0000
0	fixed $\lambda = \frac{1}{2}$	0.0268	0.0	0.0751	0.0	-0.0293	0.0158	0.0000
1	EB shrinkage	0.0545	17.7	0.0897	1.3	-0.0152	0.0225	0.0004
1	full pooling	0.1097	51.9	0.0999	3.4	-0.0049	0.0446	0.0039
1	fixed $\lambda = \frac{1}{2}$	0.0538	17.2	0.0896	1.3	-0.0153	0.0223	0.0004
4	EB shrinkage	0.0846	30.7	0.0973	1.8	-0.0079	0.0143	0.0007
4	full pooling	0.4291	92.6	0.1087	2.6	+0.0035	0.0705	0.0329
4	fixed $\lambda = \frac{1}{2}$	0.1333	61.1	0.1011	3.2	-0.0040	0.0353	0.0052
16	EB shrinkage	0.0987	40.6	0.1014	2.4	-0.0030	0.0078	0.0006
16	full pooling	1.7324	99.5	0.1064	0.4	+0.0019	0.1305	0.1628
16	fixed $\lambda = \frac{1}{2}$	0.4597	93.9	0.1072	2.2	+0.0028	0.0652	0.0359

Table 7: Three operators under the same check ($\alpha = 0.1$, $m = 12$, target from the hierarchy). Reversion caps the harm of the bad operators (full pooling, fixed λ) at the same level as that of the good one.

the “useful” region depends on the split fraction and on the dimension through the ratio of the bound to the reference risk, $\sqrt{n_e/m}/d$ up to constants, and would move in another design. The unguaranteed variants are reported, not analysed. Nothing is said about real data.

Next steps for the research line. (a) A finite-sample bound for the refit variant, in which the held-out sample enters both the check and the final estimate, or for a cross-fitted version; this is where the split cost could be recovered with a guarantee. (b) A selection bound for SURE reversion with the affine operator, whose selection region is a ball around $\hat{\mu}$ and whose risk is not given by SURE because of the discontinuity on the sphere. (c) If the original operator definitions of OMR001 are recovered, place each in Table 1 and compute $\mathbb{E}\|C - R\|$ for it; by Theorem 4.2 that is all the safety analysis needs. (d) Decide whether the line should be closed with this note or continued only along (a)–(b).

A Reproducibility

The folder contains `experiments/safe_reversion.py` (all sweeps, constants, identity re-checks, tables and figures; `-fast` runs a reduced version) and `experiments/make_numbers.py`, which converts `results/results.json` into the macros and table bodies of this document, so that every number in the text is generated and none is typed. The full run used seed 20260930 (re-checks: 20260930 + 1 and 20260930 + 1000 + i), Python 3.11.15, NumPy 2.4.6 [8], SciPy 1.17.1 [18] and Matplotlib 3.11.2, and took 17 seconds on a shared four-core machine.

α	$z_{1-\alpha}$	$\varphi(z_{1-\alpha})$	$\kappa(\alpha)$	u^*	κ/φ	$z_{1-\alpha} + \varphi(0)$
0.5	0.000	0.3989	0.1700	0.752	0.426	0.399
0.2	0.842	0.2800	0.0451	0.543	0.161	1.241
0.1	1.282	0.1755	0.0188	0.465	0.107	1.680
0.05	1.645	0.1031	0.0082	0.413	0.079	2.044
0.01	2.326	0.0267	0.0013	0.337	0.049	2.725

Table 8: Constants of Theorem 4.2 and Proposition 4.4: $\kappa(\alpha) = \sup_{u \geq 0} u\Phi(-(z_{1-\alpha} + u))$, attained at u^* , computed by bounded scalar maximisation.

Structure sweep: target drawn from the hierarchy (EB operator, $m = 12$)

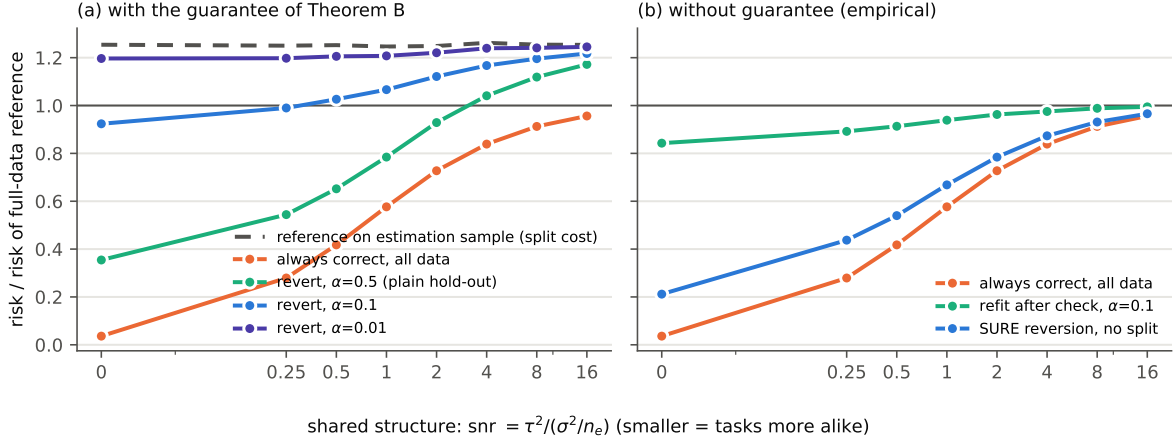


Figure 1: Structure sweep (Table 2), risks relative to the full-data reference. (a) The reverting estimators start from the split cost (dashed) and recover part of the available gain; (b) the variants without guarantee.

Draws. Per replicate: source parameters $\theta_s \sim N(0, \tau^2 I_d)$ and source means $\bar{X}_s \sim N(\theta_s, \sigma^2 I_d/n_s)$; target parameter $\theta \sim N(\kappa\tau e_1, \tau^2 I_d)$; $\bar{X}_e \sim N(\theta, \sigma^2 I_d/n_e)$ and $\bar{Y} \sim N(\theta, \sigma^2 I_d/m)$ independent; $\bar{X}_n = (n_e \bar{X}_e + m \bar{Y})/n$. The centre $\mu = 0$ without loss of generality, since the operators estimate it. Common random numbers are used across α and operators within a configuration and across m nothing is shared.

Statistics. Risks are Monte Carlo means with standard errors from the replicate standard deviation; relative gains use paired differences and treat the denominator as fixed (its relative standard error is below 1%). The Rao–Blackwellised excess $\mathbb{E}[\Delta\pi]$ uses $\pi = \Phi(-(\Delta + zs)/s)$ with the true θ of the replicate. $\kappa(\alpha)$ is computed by `scipy.optimize.minimize_scalar` on $[0, 20]$ with tolerance 10^{-10} .

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Departure sweep: negative transfer and its cap (EB operator, $m = 12$)

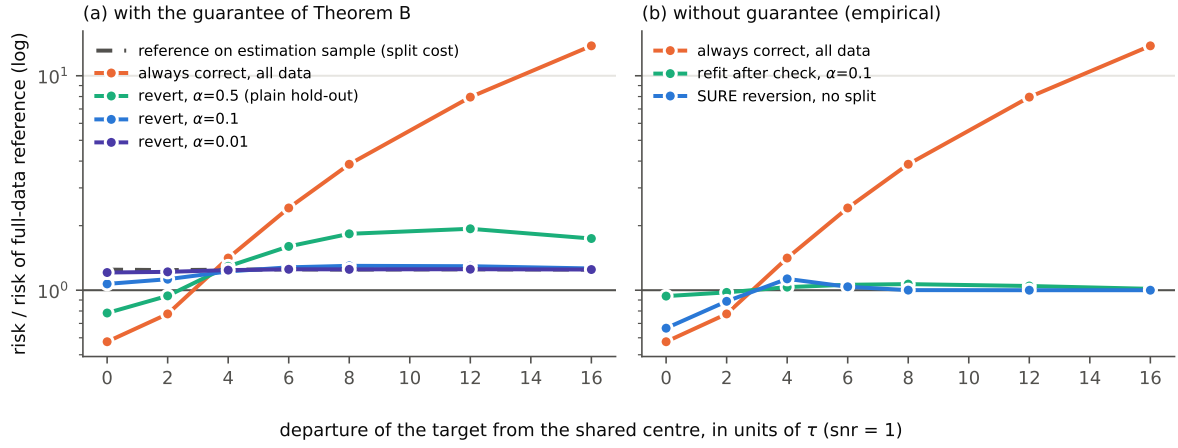


Figure 2: Departure sweep (Table 4), log scale. Always correcting is harmful from $\kappa = 4$ on; every reverting estimator stays within a bounded factor of the reference.

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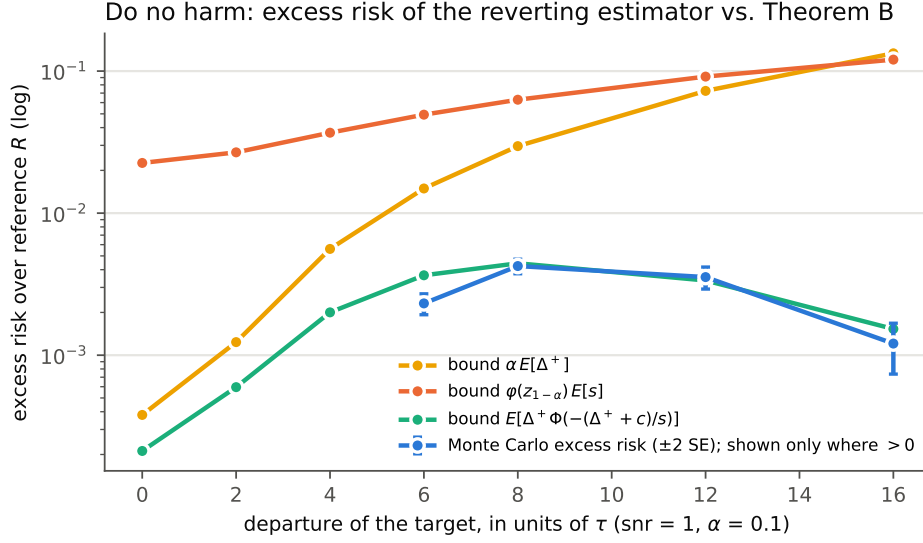


Figure 3: Excess risk of the reverting estimator over R on the departure sweep ($\alpha = 0.1$) against the three bounds of Theorem 4.2. The identity bound $E[\Delta^+ \pi]$ is nearly attained where the excess is positive; the closed-form bounds are conservative by one to two orders of magnitude here.

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claim	status
No estimator using source data whose law is free of θ beats the minimax risk $d\sigma_e^2$ (Thm. 3.1(a))	classical, proof included
A fixed additive correction is worse by exactly $\mathbb{E}\ b\ ^2$ at every θ (Thm. 3.1(b))	proved here
For $d \leq 2$, any rule that ever uses a correction is worse than R somewhere (Thm. 3.1(c))	literature + proved here
For $d \geq 3$ pointwise improvement exists but not in the worst case (Thm. 3.1(d))	literature
Bayes and frequentist risk of affine shrinkage; available gain $1/(1 + \text{snr})$ (Rem. 3.2)	classical
Held-out loss difference is conditionally unbiased and Gaussian with sd $2\sigma\ C - R\ /\sqrt{m}$ (Lem. 4.1)	proved here
Excess risk identity $\mathbb{E}[\Delta\pi]$ (Thm. 4.2(i))	proved here
checked replicate by replicate in 345 combinations	verified
Harm $\leq \min\{\alpha\mathbb{E}\Delta^+, \varphi(z)2\sigma\mathbb{E}\ C - R\ /\sqrt{m}\}$; harmful event $\leq \alpha$ (Thm. 4.2(ii))	proved here
all bounds hold in every configuration (S)	verified
Bounded-observation version via Hoeffding (Thm. 4.2(iii))	proved here
Near-oracle selection within $(z + \varphi(0))2\sigma\mathbb{E}\ C - R\ /\sqrt{m}$ (Thm. 4.2(iv))	proved here / verified
Total cost relative to full-data reference = split cost + controlled term (Cor. 4.3)	proved here
Constant sharp up to $\varphi(z)/\kappa(\alpha)$; harm not $o(m^{-1/2})$ uniformly (Prop. 4.4)	proved here; constants verified
Reverting estimator = penalised hold-out selection between two candidates (Prop. 5.1)	proved here
The line's closing statement (no differentiation from existing constructs)	supported by Sec. 5
Useful transfer (criterion U) only at $\text{snr} \leq 2$ (plain hold-out) or $\text{snr} = 0$ ($\alpha = 0.1$) in this design	verified, design-specific
Refit and SURE reversion are useful over a wider range and capped in this design	empirical only; no guarantee here
General accuracy improvement, transfer guarantee, statements about real populations	not claimed

Table 9: Claims and their status. “Verified” means reproduced by the script of Appendix A with the fixed seed; it is not a proof.